

Chapter 15: Outcome Regression and Propensity Scores

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This chapter covers **outcome regression** and **propensity score** methods (stratification, standardization, and matching on the propensity score): the most commonly used parametric methods for causal inference. They work well for treatments fixed at a single time, but they are not designed for the complexities of time-varying treatments, which is why the book presents them only after the g-methods.

This chapter is based on Hernán and Robins (2020, chap. 15, pp. 199-208).

Key theme: outcome regression and propensity score methods need the same identifying conditions as every other method (exchangeability, positivity, and consistency / well-defined interventions), plus correct models for their nuisance parameters. Unlike the g-methods of Chapters 12-14, they have limited applicability to complex longitudinal data (Hernán and Robins 2020, 199).

1 15.1 Outcome Regression (pp. 199-201)

1.1 A Structural Model with Parameters for L

Structural nested models (Chapter 14) contain parameters for A and for $A \times L$ product terms only, so g-estimation is agnostic about the L - Y relation. If instead we are willing to specify the L - Y relation within levels of A , we can use the structural model

$$E[Y^{a,c=0} | L] = \beta_0 + \beta_1 a + \beta_2 aL + \beta_3 L$$

where β_2 and β_3 are vector parameters.

In Chapter 12 this model was called a **faux marginal structural model**: it has the form of a marginal structural model, but because it conditions on the entire vector L (not a subset V), the stabilized IP weights $SW^A(L)$ all equal 1 and no weighting is needed (Hernán and Robins 2020, 199).

β_3 is often called the “main effect” of L , but it need not have a causal interpretation (the L - Y relation may itself be confounded); it only describes how the mean of $Y^{a=0,c=0}$ varies with L .

1.2 Conditional Effects from the Structural Model

Proposition 1.1 (Stratum-Specific Effect). *Under the structural model, the average causal effect in stratum $L = l$ is $\beta_1 + \beta_2 l$.*

i Proof

Proof. Evaluate the model at $a = 1$ and $a = 0$ and subtract:

$$\begin{aligned} E[Y^{a=1,c=0} | L = l] - E[Y^{a=0,c=0} | L = l] \\ &= (\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot 1 \cdot l + \beta_3 l) - (\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 0 \cdot l + \beta_3 l) \\ &= (\beta_0 + \beta_1 + \beta_2 l + \beta_3 l) - (\beta_0 + \beta_3 l) \\ &= \beta_1 + \beta_2 l. \end{aligned}$$

□

So the stratum-specific effects depend only on (β_1, β_2) ; the counterfactual means under no treatment depend on (β_0, β_3) .

1.3 Estimating the Structural Model by Outcome Regression

Under exchangeability, positivity, and well-defined interventions,

$$E[Y^{a,c=0} | L = l] = E[Y | A = a, C = 0, L = l],$$

so the structural parameters can be estimated by ordinary least squares, fitting the outcome regression model

$$E[Y | A, C = 0, L] = \alpha_0 + \alpha_1 A + \alpha_2 AL + \alpha_3 L.$$

If L suffices to adjust for confounding and selection bias and the outcome model is correctly specified, then $\alpha = \beta$.

Like stratification (Chapter 3), outcome regression adjusts for confounding by estimating the effect **within strata of L** . In Section 13.2, the outcome regression was an intermediate step toward a standardized (marginal) mean; here the regression is the end of the procedure: we compare conditional means instead of standardizing them (Hernán and Robins 2020, 200).

The same approach works for discrete outcomes, e.g., a logistic model for $\Pr[Y = 1 \mid A = a, C = 0, L]$ when Y is dichotomous.

Example 1.1 (Smoking Cessation and Weight Gain (NHEFS)). Using the outcome model of Section 13.2, which has a single product term (between A and smoking intensity), the book reports $\hat{\beta}_1 = 2.6$ and $\hat{\beta}_2 = 0.05$ (Hernán and Robins 2020, 200).

- Estimated effect $\hat{E}[Y \mid A = 1, C = 0, L] - \hat{E}[Y \mid A = 0, C = 0, L]$: 2.8 kg (95% CI: 1.5, 4.1) for those smoking 5 cigarettes/day; 4.4 kg (95% CI: 2.8, 6.1) for those smoking 40 cigarettes/day.
- Model with **no** product terms (assumes no effect modification by any variable in L): $\hat{\beta}_1 = 3.5$ kg (95% CI: 2.6, 4.3), which then estimates both the conditional and the marginal effect.

By Proposition 1.1, each stratum-specific estimate is $\hat{\beta}_1 + \hat{\beta}_2 \times (\text{cigarettes/day})$. The book reports the coefficients rounded to one or two decimals, so plugging the rounded values into that formula reproduces the reported effect estimates only approximately (e.g., $2.6 + 0.05 \times 40 = 4.6$ versus the reported 4.4).

i Fine Point 15.1: Nuisance Parameters

- **Outcome regression** for β_1, β_2 : consistent (in general) only if $\beta_0 + \beta_3 L$ correctly models how $E[Y^{a=0, c=0} \mid L]$ depends on L . The nuisance parameters are β_0 and β_3 .
- **G-estimation** of the structural nested model $E[Y^{a, c=0} - Y^{a=0, c=0} \mid L] = \beta_1 a + \beta_2 a L$: consistent (in general) only if the treatment model for $\Pr[A = 1 \mid L]$ is correct. The nuisance parameters are those of the treatment model, e.g., α_0, α_1 in $\logit \Pr[A = 1 \mid L] = \alpha_0 + \alpha_1 L$.

Example: if L should enter as $\beta_3 L + \beta_4 L^2$ but is modeled as $\beta_3 L$ only, outcome regression is biased; g-estimation is unaffected by that error but is biased if the L - A relation is misspecified. For a time-fixed treatment, choosing between the two amounts to deciding which nuisance model we can specify more accurately. When possible, a better option is a doubly robust method (Fine Point 13.2) (Hernán and Robins 2020, 200).

2 15.2 Propensity Scores (pp. 201-202)

Definition 2.1 (Propensity Score). For a dichotomous treatment A , the **propensity score** is the conditional probability of treatment given the covariates:

$$\pi(L) \stackrel{\text{def}}{=} \Pr[A = 1 \mid L].$$

- In an ideal randomized trial assigning half the individuals to $A = 1$, $\pi(L) = 0.5$ for everyone, whatever L is.
- In an observational study, $\pi(L)$ is unknown and must be estimated, e.g., with the same logistic model used for IP weighting and g-estimation in Chapters 12 and 14.

The book restricts attention to dichotomous treatments: propensity score methods other than IP weighting, g-estimation, and related doubly robust estimators are difficult to generalize to non-dichotomous treatments (Hernán and Robins 2020, 201).

Example 2.1 (Estimated Propensity Scores in NHEFS). From the logistic model for quitting smoking given L (Hernán and Robins 2020, 201):

- lowest estimated propensity score: 0.053 (individual 22941);
- highest: 0.793 (individual 24949);
- mean estimated propensity score: 0.312 among quitters ($A = 1$), 0.245 among nonquitters ($A = 0$).

Figure 15.1 of the book shows the two distributions; they overlap over most of the range.

If the distribution of $\pi(L)$ were the same in the treated and the untreated, there would be no confounding by L (no open path from L to A). Quitters having a higher average estimated probability of quitting is exactly what we expect when L predicts quitting.

2.1 The Balancing Property

Individuals with the same $\pi(L)$ may differ in L (e.g., smoking intensity and exercise), but among all individuals with a given value of $\pi(L)$ in the super-population, the distribution of L is the same in the treated and the untreated:

$$A \perp\!\!\!\perp L \mid \pi(L).$$

We say the propensity score **balances** the covariates.

Two caveats from the book (Hernán and Robins 2020, 201–2):

- The propensity score balances only the **measured** covariates L ; unlike randomization, it does nothing about unmeasured confounders.
- In a finite study population the **true** propensity score balances L only approximately, because of sampling variability. An estimated propensity score from a correct model generally gives better balance; in a randomized experiment, adjusting for the estimated $\pi(L)$ corrects both systematic and chance imbalances, whereas the true $\pi(L)$ ignores the chance imbalances.

Definition 2.2 (Balancing Score and Prognostic Score (Technical Point 15.1)).

- A **balancing score** $b(L)$ is any function of L such that $A \perp\!\!\!\perp L \mid b(L)$. The propensity score is the simplest example.
- A **prognostic score** $s(L)$ is a function of L such that $Y^{a=0} \perp\!\!\!\perp L \mid s(L)$.

Rosenbaum and Rubin (1983) proved that exchangeability and positivity given L imply exchangeability and positivity given any balancing score (Hernán and Robins 2020, 202). Graphically, $\pi(L)$ sits between L and A , with a deterministic arrow $L \rightarrow \pi(L)$ (book Figure 15.2); since $\pi(L)$ blocks every backdoor path between A and L , adjusting for it suffices.

Methods based on prognostic scores need stronger assumptions and do not extend readily to time-varying treatments; see Hansen (2008) and Abadie et al. (2013), as cited in Hernán and Robins (2020, 202).

Theorem 2.1 (Exchangeability and Positivity Given the Propensity Score).

1. If $Y^a \perp\!\!\!\perp A \mid L$, then $Y^a \perp\!\!\!\perp A \mid \pi(L)$.
2. Positivity holds within levels of $\pi(L)$ (no individual has $\pi(L)$ equal to 0 or 1) **if and only if** positivity holds within levels of L .

Hence, if L is sufficient to adjust for confounding and selection bias, so is $\pi(L)$ (Hernán and Robins 2020, 202). Positivity given the propensity score means $\Pr[A = a \mid \pi(L) = s] > 0$ for every s with $\Pr[\pi(L) = s]$ nonzero (Hernán and Robins 2020, 204).

2.2 Supplement: An Algebraic Proof of the Balancing Results

The book argues graphically (Technical Point 15.1); the same results follow from iterated expectations.

i Proof (Proof that $A \perp\!\!\!\perp L \mid \pi(L)$)

Proof that $A \perp\!\!\!\perp L \mid \pi(L)$. Since $\pi(L)$ is a function of L , conditioning on $(L, \pi(L))$ is the same as conditioning on L :

$$\Pr[A = 1 \mid L, \pi(L)] = \Pr[A = 1 \mid L] = \pi(L).$$

Conditioning on $\pi(L)$ alone, by the law of iterated expectations:

$$\begin{aligned} \Pr[A = 1 \mid \pi(L)] &= E[\Pr[A = 1 \mid L] \mid \pi(L)] \\ &= E[\pi(L) \mid \pi(L)] \\ &= \pi(L). \end{aligned}$$

The two probabilities are equal, so A does not depend on L once $\pi(L)$ is fixed: $A \perp\!\!\!\perp L \mid \pi(L)$. $\square$

i Proof (Proof of part 1 of Theorem 2.1)

Proof of part 1 of Theorem 2.1.

$$\begin{aligned} \Pr[A = 1 \mid Y^a, \pi(L)] &= E[\Pr[A = 1 \mid Y^a, L] \mid Y^a, \pi(L)] && \text{(iterated expectations; } \pi(L) \text{ is a function of } L) \\ &= E[\Pr[A = 1 \mid L] \mid Y^a, \pi(L)] && (Y^a \perp\!\!\!\perp A \mid L) \\ &= E[\pi(L) \mid Y^a, \pi(L)] && \text{(definition of } \pi(L)) \\ &= \pi(L), \end{aligned}$$

which does not depend on Y^a ; hence $Y^a \perp\!\!\!\perp A \mid \pi(L)$. $\square$

2.3 Using the Propensity Score

Under exchangeability and positivity given $\pi(L)$, we can estimate causal effects by

- stratification (including outcome regression) on $\pi(L)$,
- standardization over $\pi(L)$,
- matching on $\pi(L)$.

First estimate $\pi(L)$ from the data, then use the estimate in place of L .

3 15.3 Propensity Stratification and Standardization (pp. 202-204)

Under the identifying conditions, for each value s of the propensity score,

$$E[Y^{a=1, c=0} \mid \pi(L) = s] - E[Y^{a=0, c=0} \mid \pi(L) = s] = E[Y \mid A = 1, C = 0, \pi(L) = s] - E[Y \mid A = 0, C = 0, \pi(L) = s].$$

But $\pi(L)$ is generally continuous, so almost no two individuals share a value s (e.g., only one NHEFS individual had an estimated $\pi(L)$ of 0.6563).

3.1 Stratifying on Deciles

Example 3.1 (Decile Stratification in NHEFS). Classify individuals into 10 strata (deciles) of the estimated $\pi(L)$, about 162 individuals each, and estimate the effect within each decile (Hernán and Robins 2020, 203):

- decile-specific effects of smoking cessation on weight gain ranged from 0.0 to 6.6 kg, with wide 95% CIs;
- the same estimates come from an outcome regression of Y on A , 9 decile indicators, and 9 $A \times$ indicator product terms;
- without product terms (no effect modification by $\pi(L)$, as most applications assume): 3.5 kg (95% CI: 2.6, 4.4).

See Fine Point 15.2 (Section 15.5) for effect modification by the propensity score.

3.2 A Problem with Categorized Propensity Scores

Within a decile, the distribution of the continuous $\pi(L)$ may still differ between treated and untreated (e.g., higher average $\pi(L)$ in the treated), so the groups may not be exchangeable within that decile.

Remedy: use the estimated $\pi(L)$ as a **continuous** covariate in the outcome model $E[Y \mid A, C = 0, \pi(L)]$. In NHEFS (linear in $\pi(L)$): 3.6 kg (95% CI: 2.7, 4.5).

IP weighting and g-estimation never had this problem, because they used the numerical value of the estimated probability rather than a categorized version (Hernán and Robins 2020, 203).

Caution: for a dichotomous treatment the IP weight denominator is not $\pi(L)$ itself but a function of it: $\pi(L)$ for the treated ($A = 1$) and $1 - \pi(L)$ for the untreated ($A = 0$).

The outcome model now requires a correct specification of the $\pi(L)$ - Y relation. Because $\pi(L)$ is one-dimensional, this is easy to protect with flexible terms such as cubic splines; IP weighting and g-estimation were agnostic about that relation. Note, though, that $\pi(L)$ itself must still be estimated from a model of A on the high-dimensional L , as in IP weighting and g-estimation.

3.3 Standardizing over the Propensity Score

The outcome model on $\pi(L)$ estimates effects **within** levels s of $\pi(L)$. For the population average causal effect $E[Y^{a=1, c=0}] - E[Y^{a=0, c=0}]$, standardize the fitted conditional means over the distribution of the estimated $\pi(L)$: the Chapter 13 procedure with $\pi(L)$ in place of L .

$$\hat{E}[Y^{a, c=0}] = \frac{1}{n} \sum_{i=1}^n \hat{E}[Y \mid A = a, C = 0, \pi(L) = \hat{\pi}(L_i)]$$

Example 3.2 (Standardized Estimate in NHEFS). Standardizing over the estimated propensity score (the outcome model may include an $A \times \pi(L)$ product term) gives 3.6 kg (95% CI: 2.7, 4.6) (Hernán and Robins 2020, 203).

4 15.4 Propensity Matching (pp. 204-205)

Matching on $\pi(L)$ is analogous to matching on a single continuous variable (Chapter 4). All versions try to build a **matched population** in which treated and untreated have the same distribution of $\pi(L)$, and hence are exchangeable.

- E.g., pair each treated individual with one (or more) untreated individuals with the same, or a close, propensity score.
- Under exchangeability and positivity given $\pi(L)$, association measures in the matched population consistently estimate the effect measures **in the matched population**.

The matched population can be given the $\pi(L)$ distribution of the treated, of the untreated, or any other distribution (Hernán and Robins 2020, 204).

Variance estimation for matched estimators used to be an open problem; Abadie and Imbens (2006) solved it (Hernán and Robins 2020, 204).

4.1 Defining Closeness

Exact matches on a continuous $\pi(L)$ are rare, so we match on a **close** value, e.g., untreated individuals within ± 0.05 of the treated individual's estimated $\pi(L)$ (e.g., a treated individual with estimated $\pi(L) = 0.6563$ could be matched to an untreated individual with 0.6579).

Bias-variance trade-off:

- criteria too loose: the $\pi(L)$ distribution differs between matched treated and untreated, so exchangeability fails;
- criteria too tight: many individuals are excluded; approximate exchangeability, but wider confidence intervals.

4.2 Matching and Positivity

- In NHEFS, the estimated $\pi(L)$ distributions overlapped over most of the range; only 2 treated individuals had values above those of every untreated individual.
- Outcome regression on $\pi(L)$ (Section 15.3) kept them, implicitly treating the lack of comparable untreated individuals as **random nonpositivity**.
- Most matched analyses would drop them. Matching does not distinguish random from structural nonpositivity.

The book also states this count as a percentage of the study population (0.01%); that percentage is not reproduced here because it is inconsistent with the counts: 2 individuals are about 0.1% of a study population of roughly 1,600.

4.3 Who Is in the Matched Population?

- In theory, matching targets a well-characterized population: matching every treated individual and dropping unmatched untreated estimates the **effect in the treated** (Fine Point 15.2).
- In practice, treated individuals without a close match are excluded too, so the estimate refers to a subset defined by the values of the estimated $\pi(L)$ that found matches.

Restricting to, say, estimated $\pi(L) < 0.67$ yields a population that is hard to describe: “individuals do not come with a propensity score tattooed on their forehead” (Hernán and Robins 2020, 205), so transportability to other populations is hard to judge.

4.4 Restrict on Real-World Variables Instead

In NHEFS, the 2 treated individuals with estimated $\pi(L) > 0.67$ were the only participants over age 50 who had smoked for fewer than 10 years. Excluding them gives an estimate for

- smokers under 50, and
- smokers 50 and over who had smoked for at least 10 years,

which is a far more natural target population than “estimated $\pi(L) < 0.67$ ” (Hernán and Robins 2020, 205).

Using propensity scores to **find** the region of overlap can be useful, but restricting the study population to that region is a “lazy” way to secure positivity: the automatic positivity of matching must be weighed against the difficulty of assessing transportability (Hernán and Robins 2020, 205). Even if everyone’s propensity score were visible, the population could remain ill-characterized, because the same propensity score value can mean different things in different settings.

5 15.5 Propensity Models, Structural Models, Predictive Models (pp. 205-208)

Part II used two kinds of models for causal inference.

Definition 5.1 (Propensity Models and Structural Models).

- **Propensity models:** models for $\Pr[A \mid L]$, where L are the variables needed for conditional exchangeability. Their parameters are **nuisance parameters** with no causal interpretation.
- **Structural models:** models relating A to some feature (e.g., the mean) of the counterfactual Y^a , marginally or within levels of L . Their treatment parameters **do** have a causal interpretation. For continuous treatments they are often called **dose-response models**.

Propensity models have been used for matching and stratification (this chapter), IP weighting (Chapter 12), and g-estimation (Chapter 14). Their parameters lack a causal interpretation because L and A can be associated for many reasons: the association reflects the effect of L on A under the book’s Figure 7.1, but not under Figures 7.2 and 7.3 (Hernán and Robins 2020, 205).

5.1 Two Classes of Structural Models

- **Marginal structural models:** parameters for treatment, for chosen effect modifiers V , and for $A \times V$ product terms. V reflects only the investigator’s interest in effect modification (Section 12.5).
 - No V : truly marginal.
 - $V = L$ (all covariates): a **faux** marginal structural model, which is what outcome regression (Section 15.1) estimates.
- **Structural nested models:** parameters for treatment and for $A \times L$ product terms for those variables in L that are effect modifiers.

See Fine Point 14.1 of the book for the relation between structural nested models and faux semiparametric marginal structural models (Hernán and Robins 2020, 206).

i Fine Point 15.2: Effect Modification and the Propensity Score

- Matched and unmatched estimates may differ because of **effect modification**: when untreated far outnumber treated, matching keeps nearly all treated and drops many untreated, so the matched population resembles the treated and the estimate approaches the **effect in the treated** (Technical Point 4.1 gives IP weighting and standardization alternatives for that effect).
- Effect modification across propensity strata may suggest that decision makers treat those likely to benefit (Kurth et al. 2006, as cited in Hernán and Robins (2020, 206)), but statements like “beneficial for propensity scores between 0.11 and 0.93” have little policy

- relevance because they are not expressed in terms of measured variables L .
- Other reasons matched and overall estimates may differ: positivity violations among the unmatched, or an unmeasured confounder that is more (or less) prevalent, or better (or worse) measured, in the matched population; apparent effect modification may reflect different residual confounding across propensity strata.

5.2 Causal Versus Predictive Uses of Outcome Regression

Outcome regression is also widely used for **prediction**:

- online retailers predicting who will buy;
- clinical algorithms identifying patients at high risk.

There, no variable is “treatment” and the parameters need not have any causal interpretation.

Example from the book’s margin: Facebook Likes predict sexual orientation, political views, and personality traits (Kosinski et al. 2013, as cited in Hernán and Robins (2020, 206)); that is prediction, not necessarily causation.

5.3 Variable Selection: Prediction Tools Misapplied

Forward selection, backward elimination, stepwise selection, and machine learning are built to **improve prediction**. Applied to propensity or structural models, they can add superfluous or harmful covariates, causing

- inflated variances, and
- greater bias.

The mistaken belief behind this: that a propensity model should predict A as well as possible. It need not; it only needs to include the L that guarantee exchangeability.

Reporting predictive measures such as Mallows’s C_p for propensity models is common, but their relevance for causal inference is questionable (Hernán and Robins 2020, 207). Covariates strongly associated with A but not needed for exchangeability do not reduce bias; including them can yield very large variances or even amplify existing bias.

Example 5.1 (Hospitals Aceso and Panacea). Everyone attends hospital Aceso, where 99% receive $A = 1$, or hospital Panacea, where 99% receive $A = 0$. Hospital affects the outcome only through A , so it is not needed for exchangeability (Hernán and Robins 2020, 207).

- Adding Hospital to the propensity model gives $\pi(L) \approx 0.99$ in Aceso and ≈ 0.01 in Panacea in the target population.
- By chance, some L -strata of the study population may then contain only treated (Aceso) or only untreated (Panacea) individuals.
- Result: a near-infinite variance with **no** reduction in confounding. That A is now well predicted is irrelevant.

In the limit of perfect prediction, every treated individual has $\pi(L) = 1$ and every untreated individual has $\pi(L) = 0$: there is no overlap and the analysis is impossible (Hernán and Robins 2020, 207).

5.4 Self-Inflicted Bias and Model Specification

- Including **colliders** can cause systematic bias, even though colliders may be good predictors.
- Including **instruments** (Chapter 16) can amplify bias from unmeasured confounders.
- Chapter 18 returns to these issues.

Both propensity and structural models require a correctly specified functional form for the covariates (flexible terms such as cubic splines help), plus exchangeability, positivity, and well-defined interventions. Chapter 16 presents a method that does not require exchangeability of treatment.

6 Summary

Method	Model used	Requires
Outcome regression (faux MSM)	$E[Y \mid A, C = 0, L]$	Correct outcome model (nuisance β_0, β_3)
G-estimation (SNM)	$\Pr[A = 1 \mid L]$	Correct treatment model
Propensity stratification / outcome regression on $\pi(L)$	$\pi(L)$ and $E[Y \mid A, C = 0, \pi(L)]$	Correct propensity model and correct $\pi(L)$ - Y relation
Propensity standardization	as above	as above; targets the population effect
Propensity matching	$\pi(L)$	Correct propensity model; estimate refers to the matched population

NHEFS estimates of the effect of smoking cessation on weight gain (kg):

- outcome regression, no product terms: 3.5 (2.6, 4.3);
- propensity deciles, no product terms: 3.5 (2.6, 4.4);
- continuous propensity score: 3.6 (2.7, 4.5);
- standardized over the propensity score: 3.6 (2.7, 4.6).

Takeaways (Hernán and Robins 2020, 199–208):

- Outcome regression and propensity score methods need the same identifying conditions as the g-methods, plus correct nuisance models.
- The propensity score balances measured covariates only.
- Matching guarantees positivity automatically but can produce a target population that is hard to characterize; restricting on real-world variables is preferable.
- Propensity models should include the variables needed for exchangeability, not the best predictors of treatment.
- Unlike the g-methods, these methods do not extend readily to time-varying treatments (Part III).

Hernán, Miguel A, and James M Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC. <https://miguelhernan.org/whatifbook>.